Lesson 3.3: Modern JavaScript (ES6+)

What you learn

- **Async/await.** Writes promise code in a readable order. Always handle errors with try/catch.
- **Fetch API.** fetch returns a promise; check response.ok because HTTP errors do not reject.
- **Event loop.** Synchronous code runs first, then microtasks (promises), then timers and other tasks.
- **Modules.** import and export split code into files with clear dependencies.

Tools and resources (official sites, not checked live)

- MDN JavaScript

ASSIGNMENT

Practice for lesson 3.3

Business problem: Predict the order, then run it.

Steps

1. Read the code below.
2. Write down the output order.
3. Run it with Node.

Expected output: A to D in the order shown below.

SOLUTION (ANSWER KEY)

JAVASCRIPT

```
const order=[];
order.push('A: sync start');
setTimeout(()=>{order.push('D: timer (macrotask)');console.log(order.join('\n'));},0);
Promise.resolve().then(()=>order.push('C: promise (microtask)'));
(async()=>{await null; order.push('C2: after await (microtask)');})();
order.push('B: sync end');
```

Output order: A (sync), B (sync), C (promise microtask), C2 (after await), D (timer). Microtasks run before the timer even with a zero delay.

Check yourself

1. Why check response.ok?
   Answer: fetch does not reject on 404 or 500.
2. Which runs first: a promise callback or setTimeout 0?
   Answer: The promise callback.

Lesson 3.3 of the **Development and IT** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on

each official site.